

The ModelSEED Biochemistry Database, 2026 update: grading multi-source thermodynamics

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Abstract

The ModelSEED Biochemistry Database (<https://modelseed.org>) supplies foundational mass- and charge-balanced reaction networks for metabolic reconstructions. Here we present an update on the biochemistry where the database has been expanded to ~ 46,000 compounds, and ~ 48,000 reactions, featuring ~ 37,000 metabolic structures. We have expanded our approach for handling thermodynamic data, enabling multiple sources of data to be derived, integrated, and presented to the wider research community. We now publish predictions of pKa, reaction energy (and respective uncertainties), and estimates of reaction direction from multiple independent sources. Reaction directions are graded gold, silver, or bronze according to the strength of supporting evidence, and are complementarily predicted by an ensemble of large language models. These diverse sources empower users to understand agreements or discrepancies in the directionality for ~ 27,000 reactions. Finally, to ensure data integrity, a new conflict-resolution pipeline reconciles structures across sources, documenting input from curators. Our work is publicly available at <https://github.com/ModelSEED/ModelSEEDDatabase>.

Keywords biochemistry database, metabolic modelling, thermodynamics, reaction reversibility, ModelSEED

Introduction

The ModelSEED Biochemistry Database provides mass- and charge-balanced reaction networks reconciled across major biochemical resources for metabolic reconstructions (1). Since its 2020 release, the database has been widely adopted for curating and parameterizing genome-scale metabolic modeling and integration. For example, GEMsembler utilizes the shared ModelSEED namespace to resolve cross-tool identifier mappings (2), models (GEMs): e.g. GEMsembler utilizes it for cross-mapping between resources (2) and Yeast9

incorporates its metabolite values to parameterize reaction Gibbs energies across the network sources Gibbs energies from it for parameterizing GEMs (3).

This update reports the growth of the database, and an expanded approach in handling the layer both expanding the database with more reaction biochemistry and a layered handling of thermodynamic sources and reaction directionalities that communicates the breadth of thermodynamics data allowing us to publish every source with its own uncertainty rather than promoting one. While over 70% of the reactions with multiple

sources of data, where confidence of reaction directionalities are expressed as a grade based on the strength of supporting data. While > 70% of sources agree on reaction direction, any disagreement is published for review by researchers to decide which numbers and approach carry weight.

Here we also describe a grading approach incorporating the uncertainties of the data across different sources, an integration of atom-mapping data directionality, disagreements are displayed for investigators to interpret. Atom-mapping data is further integrated to enable MFA analyses, and a structure-curation pipeline is built to resolve conflicts between structures from different databases. Finally, we expanded our search interface at <https://modelseed.org> for dissemination of the results to facilitate user queries of the resource.

Materials and Methods

Construction and Curation

We collated additional compounds and reactions from MetaCyc (4), Rhea (5), and ChEBI (6). The compounds are reconciled into single ModelSEED compounds using available merging compounds via structures where possible and the reactions are reconciled based on the identity of the compounds. The procedure follows the matching reactions by compound identities, similar to the 2020 release (1). Mass and charge balance is enforced and flagged.

Unlike the 2020 release, a conflict between structures from multiple sources was where compounds with conflicting sourced structures were either manually reviewed or dropped silently. The updated pipeline classifies each structural conflict and routes it to one of three outcomes: an automatic pick structural conflicts and either 1) automatically picks from the source variant matching that matches the historical formula, a curator-authored override with recorded rationale, or an entry in a mass-balance exclusion registry for compounds where no source formula is defensible. Every 2) prioritizes curator authored and justified content, or 3) exclusion (with logged justification) for compounds without a defensible source formula. This enforces that every outcome is attributed to its curator and its evidence. We retained every structure we collect, so conflict resolution can be inspected within the data evidence. Every structure is retained in the repository. An outline of how a researcher can contribute is given to facilitate conflict resolution. Investigators can contribute to the ModelSEED Database through the procedure in Supplementary Methods S1 and, which is expanded in the repository.

Multi-source thermodynamics

The 2020 release combined two sources: the historical group contribution approach (7) and eQuilibrator (8). This update carries four sources, each stored as its own record: group contribution, thermodynamics sources: eQuilibrator (8) and the group contribution method (7). This ModelSEED Database update incorporates the most recent eQuilibrator 3.0 (9) and introduces two new thermodynamics sources: dGPredictor (10) and experimental values where they exist. Each of the four distinct thermodynamics sources are reported separately, where

available. Only the experimental values carry a published direction. The three predictors were released as energy estimators, so every direction reported here is an inference from their energies. Other three sources are energy estimates and their resulting directions are inference. All values are reported at pH 7.0, ionic strength 0.25 M, pMg 3.0 and 298.15 K; the basis for each condition and its effect on the released energies are given in Supplementary Methods S2. A key caveat: as we publish a generalized database that can be applied across the kingdoms of life, transport reactions are scored from stoichiometry and energy alone, and neither an estimate of membrane potential or pH gradients are used.

Protonation.

eQuilibrator's Legendre transform requires a macroscopic pKa ladder per compound: an ordered list in which consecutive species differ by one proton. The count of entries around the desired pH sets the proton count of the major microspecies, so the composition of the ladder changes the transformed energy directly. We generate the pKa ladders using ChemAxon Marvin (11) for use with eQuilibrator, but we're conscious that these data cannot be regenerated without a license, so we also include the pKas generated by an open-source alternative, MolGpKa (12), for every compound with a structure (source provenance preserved), and the measured ladders of Alberty (13), versioned per structure source in an extended pKa dictionary using licensed ChemAxon Marvin (11) for use with eQuilibrator and also via open-source MolGpKa (12) for unlicensed use.

Experimental anchors.

eQuilibrator and dGPredictor draw on the same set of experimental values derived from the NIST Thermodynamics of Enzyme-catalyzed Reactions (TECR) Database (14). The work was keyed to KEGG, and eQuilibrator uses MetaNetX mappings to integrate with other resources but there are inherent structural conflicts when we examine the mappings to ModelSEED identifiers. In order to draw a challenge confronted here was structural conflicts in compounds mapped to ModelSEED from the KEGG IDs of TECR and the MetaNetX IDs of eQuilibrator. To facilitate direct comparisons between the results generated for any ModelSEED reactions thermodynamic predictions of ModelSEED reactions, we re-built the experimental anchors of TECR values, curating any conflicts to ensure a direct mapping between TECR values and ModelSEED reactions. In addition we evaluated and curated conflicts to directly map TECR to ModelSEED. We further investigated openTECR, a communal re-curation of the same source, TECR available at <https://github.com/opentecr>, to see if we can extend the set of extend the thermodynamic anchors.

Uncertainty.

The three sources each adopt a different approach for estimating the uncertainty surrounding the predicted energy of reaction and as such the scale of different thermodynamic sources estimate reaction free energy uncertainties differently, and consequently uncertainty is not directly comparable (Figure 2C)) median values (kcal mol^{-1}) are 0.63 for eQuilibrator, 10.41 for group contribution, and 17.01 for dGPredictor. Comparing against the experimentally-anchored reactions, we find that group

contribution overstates its error and eQuilibrator understates it.

Direction.

For historical reasons To conserve precedence, we use the same heuristics for deriving reaction direction 2020 heuristics to derive reaction directions from the Group contribution approach (7) unchanged from 2020, so that series stays comparable across releases. For eQuilibrator and dGPredictor, we use one rule set built on the reversibility index from Noor *et al.* (15): a reaction is called irreversible when $|\ln \Gamma|$ exceeds $\ln(1000)$ by more than one propagated standard deviation (9), Γ being the fold-change of reactant concentration required to change reaction direction). However, we extend the work by Noor *et al.* to require rule to require that the uncertainty clears the uncertainty to clear the threshold as well. We report fewer directions than the published index would on the same energies, and we separate reversible reactions from undetermined cases. An exception we implement is except for reactions whose uncertainty is small relative to the threshold and yet still straddles it: meaning much smaller than the threshold yet the estimate lies within one standard deviation of the threshold, and these we report as reversible, which we call reversible.

Transport.

Compartments are relative indices — 0 inside, 1 outside — not named compartments, so one reaction can be instantiated against a bacterial membrane, a mitochondrion or a plastid envelope when a template builds a model. Because the database is not committed to a cell type, transport reactions are scored from stoichiometry and energy alone: neither membrane potential nor pH gradient enters the estimate. The consequence is specific. A primary active transporter carries ATP hydrolysis in its stoichiometry, every predictor estimates that tightly, and the reaction earns a confident grade from chemistry that was never in question — 96% of gold-graded transport reactions are ATP-coupled and only 2.7% translocate a proton. Such grades measure confidence in the coupled chemistry, not evidence that the transporter runs that way in a given organism; every reaction carries an is-transport field so that this can be accounted for downstream. Reactions whose only change is of compartment carry no net chemistry and none is graded gold. dGPredictor commits on 1.4% of transport reactions against 17.3% elsewhere, so transport grades also rest on fewer independent sources.

Grading reaction direction

We apply a classification approach with which we grade the predicted reaction direction based on the quality of available evidence. Very often sources and heuristics disagree, as is the case here, and we qualify what the reaction direction would be, and how reliable the evidence is using several tiers for ease of interpretation: gold. The selected tiers are intuitively described as gold (highest), silver (intermediate), silver, or bronze (lowest). Our evaluation splits two ways, on the confidence of a source's own claim, fitted as a probability against the experimental anchors, and what the other sources make of it, by a weighted comparison on the same scale. Grades, self-assessments, and cross-source verdicts are stored in the biochemistry database and can be viewed in the UI. The process of grading reactions

is described fully and exposed on the website for users to inspect disagreements among the sources and heuristics for themselves. The grading methodology is thoroughly described in Supplementary Methods S3.

Ensemble LLMs predictions

Thermodynamic assignment is bounded by estimate coverage: energies cannot be computed for an incomplete reaction in terms of structures. Furthermore, every reaction is a member of a pathway within a cell, and the overarching drive of the pathway, dynamically changing metabolite concentrations as downstream enzymes process reagents, means a reaction may be driven in a direction on a level that supersedes our evaluation (16; 17; 18), leaving 23,790 reactions with no direction from any predictor. We run a complementary approach using an ensemble, a "council" of large language models (LLMs) to interpret the reaction based on its name and stoichiometry alone. Three LLMs independently make their predictions, a fourth audits, and a fifth adjudicates. We provide the prompts in the repository, and describe the process and its limits in Supplementary Methods S4. We integrate these predictions transparently in our database, but do not include them in our grading of the thermodynamic evidence.

The ensemble directs 22,963 of the reactions no thermodynamic source will commit on, roughly doubling the fraction of the database that carries a direction. Its accuracy is not that of an energy calculation: scored against measurement it recovers the direction 77% of the time against 98% for eQuilibrator, and 85% of its calls are simply the direction the equation is written in, so its errors fall almost entirely on reactions that run in reverse (Supplementary Methods S4). We therefore release these calls as a separate, clearly-labelled layer, excluded from the evidence grading: they are for reactions a reconstruction would otherwise leave reversible by default, and should be reviewed before use rather than treated as evidence. Neither approach models the pathway context that can drive a reaction against its own standard-state energy as metabolite concentrations shift downstream (16; 17; 18); that remains a property of a model rather than of a reference database.

Atom mapping

Atom mappings are generated from the workflow of Huß *et al.* (19) where they refine the per-reaction output of the Reaction Decoder Tool (20) into mappings that can be used to span a metabolic reconstruction. The approach also resolves chemically equivalent atoms into symmetry groups, such as the two oxygens of CO₂ and the six equivalent terminal oxygens of pyrophosphate, so the ambiguity is apparent. The output is formatted and stored in the database, and can also be visualized in the UI.

Distribution and community process

The database is distributed as versioned flat files and JSON from the public repository, queried programmatically through a SOLR endpoint, and browsed through the web interface at <https://modelseed.org>. Every merge runs continuous integration that rebuilds the database from source records and

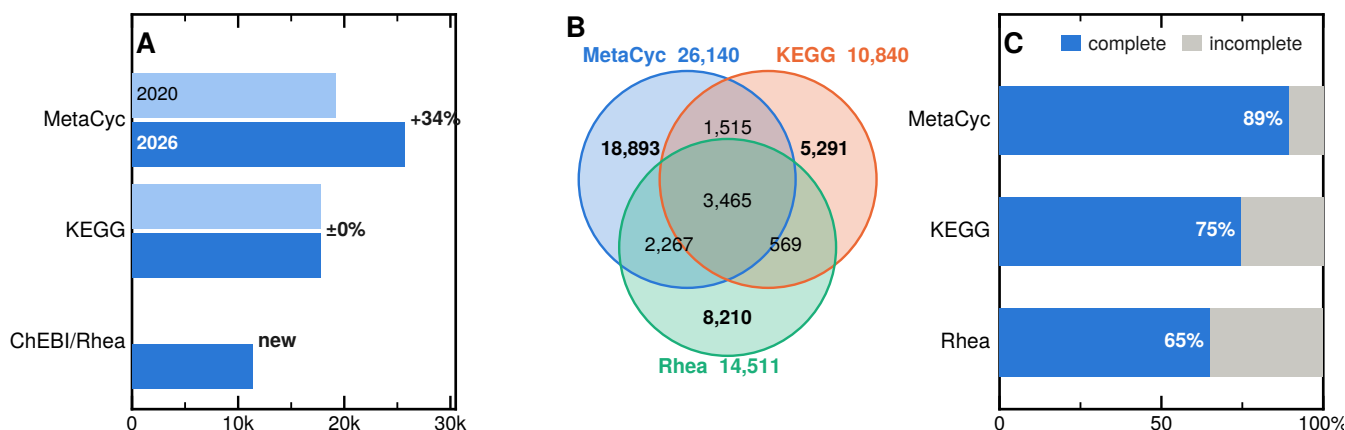


Figure 1 Sources of compounds and reactions. (A) compounds contributed by each source that also supplied molecular structures, 2020 against 2026: we did not introduce any new biochemistry from KEGG. The third row is ChEBI, which supplies structures but no reactions. (B) how the share of each three primary database's reactions that reaction sources overlap, every region labelled with its reaction count. Circle areas are unique; not proportional — three-set areas cannot be solved exactly — so the counts carry the quantities and the circles only carry membership. Rhea identifies its compounds through ChEBI. (C) the share of those each source's unique reactions that are complete, meaning every reagent in a reaction is assigned a complete structure, so the reaction can be balanced and decomposed. The three panels share one row per source; the third row is ChEBI in (A) and Rhea in (B) and (C), because ChEBI supplies structures but no reactions and Rhea supplies reactions but no structures. Rhea identifies its compounds through ChEBI, so its completeness is dependent on ChEBI structures.

fails on any regression in balance, structure coverage or schema conformance, so a curation change that breaks an invariant cannot be merged. Researchers also can contribute by adding and curating structures and reactions, the repository has a guide for doing so, and new data can be merged guidance for which is available in the repository and is managed via pull requests.

Results

Growth and coverage

The database has grown 34% in compounds and 5534% in reactions since 2020, counting only non-obsolete entries at both time points, driven mainly by MetaCyc and the BioCyc family. Reaction growth is now split across three primary databases (Figure 1B): MetaCyc carries 31,805–26,140 reactions of which 21,258–18,893 are supplied by no other primary source, KEGG 14,066–10,840 of which 5,756–291 are unique, and the newly-integrated Rhea 17,477–14,511 of which 8,339–210 are unique: reactions the database would not otherwise hold. Completeness of reactions differs (Figure 1C). We classify a complete reaction where every reagent can be assigned a complete molecular structure: 90% of MetaCyc's and 7675% of KEGG's reactions against 6665% of Rhea's. Overall structure coverage rose from 28,120–087 to 36,943–compounds–907 compounds. The new biochemistry is not distributed like the old (Supplementary Figure S1): the MetaCyc classes that grew are those of specialised metabolism — antibiotic and polyketide biosynthesis, O-antigen and cell-envelope assembly, branched and acylated lipids, alkaloids — while central carbon and energy metabolism account for 41 of the 12,259 reactions added. Central metabolism was already saturated in 2020; what a 2026 reconstruction gains is the periphery. The comparison covers the

MetaCyc-sourced additions; Rhea, which supplied 8,409 of the new reactions, has no pathway ontology to place them in.

Structure curation

We built and ran our curation pipeline on all structures that would impact the set of roughly 9,000 reactions used in the ModelSEED and PlantSEED reconstruction templates, since an error in one of those propagates into every model built from them. This scope reflects a curation priority, not a statement about how much of the database is usable: the templates are discussed in Section 3.5. The conflict pipeline resolved every previously conflicting compound in three curation rounds: i) Instances where the conflict existed at the level of mobile hydrogens were resolved automatically using historical formula ii) instances where the structures were structurally different were resolved manually iii) instances where resolution was not straight-forward were entered in the a three-staged curation: i) historical formula were used to automatically resolve conflicts in mobile hydrogens; ii) manual curation resolved conflicts in molecular structures; and iii) where steps i-ii fail because of an absence in defensible source formula, reactions were added to a mass-balance exclusion registry where we fix an arbitrarily chosen formula because no source formula was defensible.

Thermodynamics

The three predictors cover comparable fractions of the database and disagree substantially about how well they do it (Figure 2B). Uncertainty is reported on scales differing by more than an order of magnitude (Figure 2C): a median of 0.63 median errors (kcal mol⁻¹) of 0.62 for eQuilibrator, 10.83 for eQuilibrator against 10.41 kcal mol⁻¹ for group contribution and 17.01 kcal mol⁻¹ and 17.57 for dGPredictor.

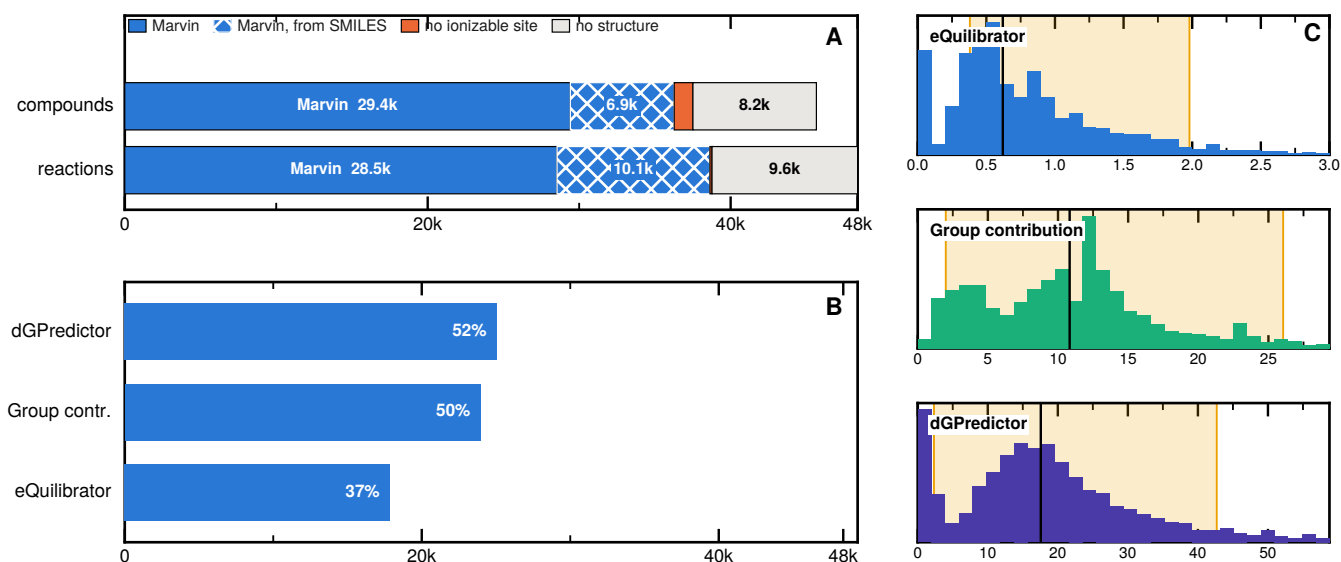


Figure 2 Thermodynamic sources. (A) Provenance of every pK_a ladder shipped in the database, across all 45,708,662 compounds and 5648,002 384 reactions. ChemAxon Marvin supplies 79.4% of compounds and 81.679.8% of reactions. Segments are identified by the key. Hatching marks the share reached through a second route: the calculator refuses polymers and organometallics as query molecules, so those are computed from SMILES instead. No ionizable site is a chemistry result — Marvin ran and reported no dissociable proton between pH −2 and 16, for 1,242 237 compounds — The remainder carry no structure to compute — and is coloured apart from no structure, which is a curation gap rather than a protonation one. (B) Reactions that carry an energy computed from the source heuristic. (C) Reported uncertainty in kcal mol⁻¹, one axis per source because the three differ by more than an order of magnitude. The shaded band spans the 5th to 95th percentile, which is used in our classification scheme.

Calibrating the reported errors against measurement.

We use the experimental anchors to evaluate the error estimates. For each source, we collect one held-out prediction per anchored reaction, together with the uncertainty that source reports for it, and divide the residual against the measured $\Delta_r G'^{\circ}$ by that uncertainty. Measured this way, eQuilibrator has a root mean square of 8.06 and a median of 2.27, with only 28.7% of reactions within one reported standard deviation and 46.3% within two, so it understates its error. Group contribution runs the other way, with, by contrast, has a median of 0.32 and 73.6% within one standard deviation, and dGPredictor is the best-scaled of the three, performs even better with a root mean square of 1.15 and 87.9% within one. A researcher choosing by smallest reported error would systematically choose the source that understates it, which is the argument against promoting any single source. standard deviation. The grading in the next section calibrates each source's confidence against measurement rather than taking its uncertainty at face value.

Direction assignment and evidence grading

Each source now generates its own prediction for the direction each reaction proceeds, and the three sources differ more in what they will commit to than in what they say (Figure 3A). eQuilibrator resolves 84.397.4% of the reactions it covers, group contribution 42.959.7% and dGPredictor 41.140.1%. Across the database, 30,157–24,594 reactions receive a direction or a positive statement of reversibility from at least one source and 25,855–23,790 receive none.

eQuilibrator vs dGPredictor

dGPredictor is able to generate a predicted energy for more reactions (~29,600~25,100, against eQuilibrator's ~21,800~17,900) and converts the least of that coverage into a usable reaction direction. For a typical dGPredictor reaction the error bar alone exceeds the decision threshold in the reversibility index, so 58.959.9% of its reactions are returned undetermined against eQuilibrator's 15.72.6%. Where both sources commit to a direction they agree 95.094.8% of the time. dGPredictor is abstaining rather than contradicting, and it resolves 4,248–3,877 reactions eQuilibrator cannot, against 13,289–11,237 the other way.

Grading.

Of 33,099–27,338 graded reactions, 3,434–4,073 are gold, 18,388 silver and 11,277–14,475 silver and 8,790 bronze (Figure 3DB,C; see classification scheme in Supplementary Methods S3). The tiers separate on what supports them: gold is 77.91% self-certain and 68.85% corroborated, while bronze is 99% unconfident and split between reactions no second source could check (48%) and reactions the other sources contradict (35.29%). Because a measured reaction is always graded gold, the grades must be recomputed with openTECR withheld before they can be tested against it. Re-graded on the predictors alone, the 806 anchored reactions split into 357 gold, 437 silver and 12 801 anchored records split into 439 gold, 349 silver and 13 bronze; the gold subset falls within 2 kcal mol⁻¹ of measurement 96.993.8% of the time, against 91.389.4% for silver. This validation counts records rather than distinct reactions: 438 of the 801 are obsolete duplicates of a live entry, so the effective number of anchored reactions is 363 and these rates

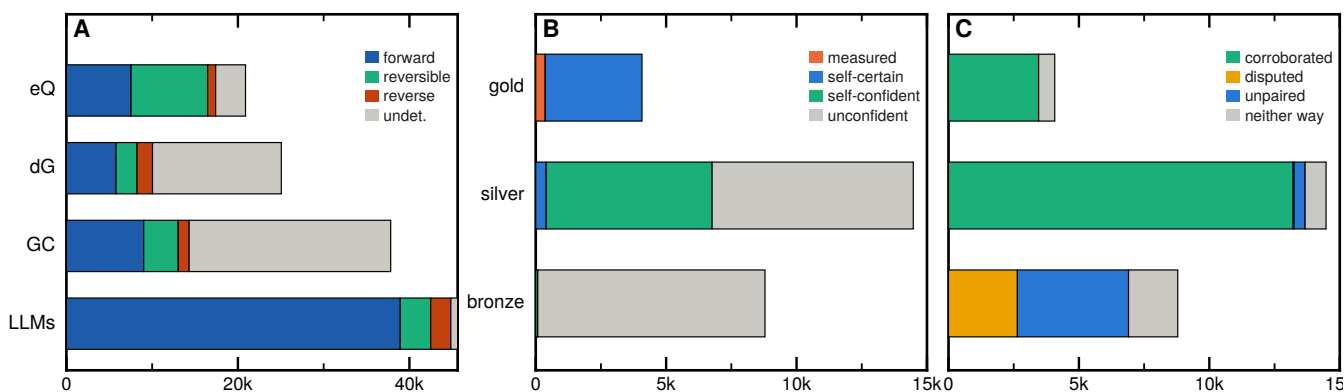


Figure 3 Direction, agreement and evidence. Direction, agreement, and evidence. (A) Direction assigned by each source independently: eQ eQuilibrator, dG dGPredictor, GC group contribution, and LLMs the large language model ensemble, which supplies a that predicts direction and no but not energy; its calls are released with the database and take no part are not considered in the evidence grading. Bar length is the number of reactions the source covers, so the rows are directly comparable facilitating comparison. (B) What each evidence grade rests on by the deciding source's own calibrated confidence, and (C) by what the other sources made of it. The two panels share a scale but not a palette, so each carries its own key; categories are ranked as in Supplementary Table 1. Neither way is the case that table writes ---: a cross-check ran and settled neither way, so it neither helped nor penalised penalized.

rest on fewer independent measurements than the counts imply (Supplementary Methods S3). Both tiers therefore track measurement closely enough that we recommend either for inclusion in a metabolic model, and reserve bronze for review rather than use.

Recommendation.

Evidently where the sources agree on reaction direction, this becomes our recommendation, but because the sources can disagree, we prioritize the source from which we take the recommended reaction direction, so the priority order is eQuilibrator, dGPredictor, group contribution. eQuilibrator supplies 21,218–16,985 recommended reaction directions, dGPredictor 4,248–7,935 and group contribution 4,691–2,055; these numbers include the ones where there is agreement also. We record the source used to make the recommendation, so researchers may apply a different precedence.

Atom mapping

Atom mappings are published for 32,877 reactions, 5926,242 reactions, 54% of the database: 25,058–19,809 clean, and 7,819 6,433 that required salvage (which are tagged). As the mappings can be used to span a metabolic reconstruction, reactions needing one repair are likely to carry others.

Reconstruction scope.

A template carries far fewer reactions than the database holds — the v7.0 ModelSEED templates span 8,597 in union, about 9,000 with PlantSEED.v3 roles — but that gap does not mean the remainder is unusable. The database is a reference namespace across all kingdoms and includes biochemistry from secondary databases and published models that no single template targets. Promotion into a template is limited by mass and charge balance, which 69% of reactions satisfy, and by association to an annotated functional role, which 41% carry. Gene association is not the constraint: 2,258 of the

8,584 reactions in the Gram-negative template are typed as gapfilling and require no gene. 15,106 reactions now satisfy both balance and role annotation, 8,815 of them outside every current template.

Discussion

Unlike the 2020 release, this update adopts a multi-source policy rather than promoting a single thermodynamic value, publishing four sources, each with their own uncertainty and estimated direction, because the sources may disagree in ways a single value conceals. Grading against experiment enables us to validate the approach: A researcher can take the highest-graded source and knows what the grade means. Two limits are worth naming. Transport reactions are scored without a membrane potential or a pH gradient, because both are properties of an organism and a condition rather than of a reference reaction; a compartment-aware scoring layer is the natural next release, and it needs the organism context a model supplies and a database cannot. And roughly half the database still carries no thermodynamic direction at all, bounded by the structures needed to compute one. In being explicit about disagreement between sources, we expect this release to serve the widening range of communities that researchers who build on shared biochemistry: genome-scale reconstruction, flux analysis, pathway design and the training of predictive models.

Data availability

The ModelSEED Biochemistry Database is available at <https://github.com/ModelSEED/ModelSEEDDatabase> under the Creative Commons Attribution License, as versioned flat files and JSON, and archived at Zenodo (<https://doi.org/10.5281/zenodo.22782602>). Data directly derived from KEGG and MetaCyc remain subject to the licences of those resources. It is also queryable through a Solr endpoint and browsable at

<https://modelseed.org>. Released structures, thermodynamic values with per-source uncertainty, evidence grades, atom mappings and the low-confidence reaction flags are all in that repository. The scripts that regenerate every published value, including the eQuilibrator cache rebuild, are under [Scripts/Thermodynamics/](#).

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Author contributions

Contributions follow the CRediT taxonomy. S.M.D.S.: conceptualization, methodology, software, formal analysis, data curation, visualization, writing – original draft, project administration. C.S.H.: conceptualization, funding acquisition, supervision. A.F., J.P.F., V.S. and C.T.: resources, data curation, investigation; J.P.F. also writing – original draft. J.N.E. and F.L.: resources, software (supporting). R.T.G.: the openTECR re-curation of the experimental measurements. E.N. and M.E.B.: eQuilibrator. V.U., M.A. and C.D.M.: dGPPredictor. S.H. and Z.N.: atom mapping. A.P.A., R.W.C. and E.M.W.-C.: funding acquisition, supervision. All authors reviewed the manuscript.

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Conflict of interest

None declared.

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References

- Seaver, S. M. D., Liu, F., Zhang, Q., et al. (2021) The ModelSEED Biochemistry Database for the integration of metabolic annotations and the reconstruction, comparison and analysis of metabolic models for plants, fungi and microbes. *Nucleic Acids Research*, **49**(D1), D575–D588.
- Matveishina, E. K., Bartmanskii, B. J., Benito-Vaquerizo, S., and Zimmermann-Kogadeeva, M. (2025) GEMsembler: consensus model assembly and structural comparison of genome-scale metabolic models across tools improve functional performance. *mSystems*, **10**(10), e0057425.
- Zhang, C., Sánchez, B. J., Li, F., et al. (2024) Yeast9: a consensus genome-scale metabolic model for *S. cerevisiae* curated by the community. *Molecular Systems Biology*, **20**(10), 1134–1150.
- Caspi, R., Billington, R., Keseler, I. M., et al. (2020) The MetaCyc database of metabolic pathways and enzymes — a 2019 update. *Nucleic Acids Research*, **48**(D1), D445–D453.
- Bansal, P., Morgat, A., Axelsen, K. B., et al. (2022) Rhea, the reaction knowledgebase in 2022. *Nucleic Acids Research*, **50**(D1), D693–D700.
- Hastings, J., Owen, G., Dekker, A., et al. (2016) ChEBI in 2016: Improved services and an expanding collection of metabolites. *Nucleic Acids Research*, **44**(D1), D1214–D1219.
- Jankowski, M. D., Henry, C. S., Broadbelt, L. J., and Hatzimanikatis, V. (2008) Group contribution method for thermodynamic analysis of complex metabolic networks. *Biophysical Journal*, **95**(3), 1487–1499.
- Flamholz, A., Noor, E., Bar-Even, A., and Milo, R. (2012) eQuilibrator — the biochemical thermodynamics calculator. *Nucleic Acids Research*, **40**(D1), D770–D775.
- Beber, M. E., Gollub, M. G., Mozaffari, D., et al. (2022) eQuilibrator 3.0: a database solution for thermodynamic constant estimation. *Nucleic Acids Research*, **50**(D1), D603–D609.
- Wang, L., Upadhyay, V., and Maranas, C. D. (2021) dGPPredictor: Automated fragmentation method for metabolic reaction free energy prediction and de novo pathway design. *PLOS Computational Biology*, **17**(9), e1009448.
- ChemAxon Marvin 26.1. <https://www.chemaxon.com> (2026) Chemical structure and pK_a prediction toolkit.
- Pan, X., Wang, H., Li, C., Zhang, J. Z. H., and Ji, C. (2021) MolGpKa: A Web Server for Small Molecule pK_a Prediction Using a Graph-Convolutional Neural Network. *Journal of Chemical Information and Modeling*, **61**(7), 3159–3165.
- Alberty, R. A. (2003) Thermodynamics of Biochemical Reactions, Wiley-Interscience, Hoboken, NJ.
- Goldberg, R. N., Tewari, Y. B., and Bhat, T. N. (2004) Thermodynamics of enzyme-catalyzed reactions — a database for quantitative biochemistry. *Bioinformatics*, **20**(16), 2874–2877.
- Noor, E., Bar-Even, A., Flamholz, A., et al. (2012) An integrated open framework for thermodynamics of reactions that combines accuracy and coverage. *Bioinformatics*,

- 28(15), 2037–2044.
16. Mavrovouniotis, M. L. (1993) Identification of localized and distributed bottlenecks in metabolic pathways. In *Proceedings of the International Conference on Intelligent Systems for Molecular Biology* PMID 7584346; no DOI registered Vol. 1, pp. 275–283.
 17. Xu, M., Smith, R., and Sadhukhan, J. (2008) Optimization of productivity and thermodynamic performance of metabolic pathways. *Industrial & Engineering Chemistry Research*, 47(15), 5669–5679.
 18. Noor, E., Bar-Even, A., Flamholz, A., et al. (2014) Pathway thermodynamics highlights kinetic obstacles in central metabolism. *PLoS Computational Biology*, 10(2), e1003483.
 19. Huß, S., Judd, R. S., Koper, K., Maeda, H. A., and Nikoloski, Z. (2022) An automated workflow that generates atom mappings for large-scale metabolic models and its application to *Arabidopsis thaliana*. *The Plant Journal*, 111(5), 1486–1500.
 20. Rahman, S. A., Torrance, G., Baldacci, L., et al. (2016) Reaction Decoder Tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13), 2065–2066.